

# Next-Gen Music Recommender on Yambda — Technical Report

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## TL;DR

We build a next-track recommender on real Yandex Music logs and evaluate it under the dataset's official protocol. A **SASRec** sequence model beats every published baseline and reproduces the paper to **98 %**. A **late-fusion hybrid** that adds an audio-content score lifts NDCG@10 by a **robust +9.7 % ± 2.0 %** over SASRec across 5 held-out user splits. A tail analysis shows the audio gain comes from **mainstream tracks, not the long tail** — overturning our initial cold-start hypothesis and giving an honest account of *why* the hybrid works.

| Model                           | NDCG@10       | Note                                         |
|---------------------------------|---------------|----------------------------------------------|
| MostPop                         | 0.0171        | popularity                                   |
| ItemKNN (bm25)                  | 0.0709        | strongest classical baseline                 |
| <b>SASRec</b>                   | <b>0.0735</b> | beats all baselines; 98 % of paper (0.0748)  |
| Hybrid — joint embedding fusion | 0.0581        | overfit → negative result                    |
| <b>Hybrid — late fusion</b>     | <b>0.0798</b> | <b>+9.7 % ± 2.0 %</b> over SASRec (5 splits) |

## 1. Problem

Given a user's listening history, predict the next track they will actually want to hear — the decision a streaming product makes for every active user, every few minutes. In a saturated market where subscriber growth has plateaued, the quality of this ranking is the lever on listening time and churn. We target a measurable improvement in offline next-track ranking.

## 2. Data

**Yambda** is a production listening log from Yandex Music. The full release has 4.79 B events / 1 M users / 9.39 M tracks; we use the official **50M** slice (the right size for a single free GPU while preserving all real-world structure and matching the published baselines).

After our preprocessing (see §4):

| Quantity                               | Value                  |
|----------------------------------------|------------------------|
| Raw listen events (50M)                | 46,467,212             |
| Listen+ events (played $\geq 50$ %)    | 29,439,278             |
| Train Listen+ (after GTS split)        | 29,285,498             |
| Test Listen+ (1-day window)            | 152,899                |
| Item vocabulary (distinct train items) | 629,373                |
| Evaluated users (have train history)   | 4,549                  |
| Audio embedding dim                    | 128                    |
| Audio coverage of our vocab            | 94.6 % (595,237 items) |

**Signal layers used.** Implicit feedback (plays with played-ratio), and the 128-d precomputed audio embeddings. Every event also carries an `is_organic` flag (user-driven vs recommender-driven) — a debiasing handle most public datasets lack.

**Real-world operational data** from a production company → qualifies for the +2 bonus.

### 3. Models

**Baselines.** MostPop (global popularity) and ItemKNN (item-based kNN via `implicit`, cosine / tfidf / bm25 weightings).

**SASRec** (sequence tower). A causal-attention Transformer over the user's Listen+ history (structurally a small GPT): item + positional embeddings, 2 self-attention blocks ( $d = 64$ , 1 head, dropout 0.2, maxlen 200), trained with the original BCE + one-negative-per-position objective for 120 epochs.

**Audio-content tower.** A training-free content score: the user is the **mean audio embedding of their history** (a "taste centroid"), items are scored by audio dot-product. It can score any item with an audio vector.

**Fusion — the key design decision.** - *Joint embedding fusion (failed).* We first folded content into the item embedding,  $v_j = e_j + \alpha \cdot \text{ContentMLP}(\text{audio}_j)$ , trained end-to-end. It **overfit** (train loss fell below SASRec's while test NDCG dropped to 0.0581): joint training lets the content branch memorise training transitions and dilute the collaborative signal. The learned  $\alpha$  stayed high ( $\sim 1.18$ ), i.e. the optimiser *wanted* content because it helped *training* — the textbook overfit signature. - *Late (score-level) fusion (won).* Keep SASRec frozen and combine standardised scores:  $\text{score}(u, j) = \text{zscore}_j(\text{SASRec}) + \beta \cdot \text{zscore}_j(\text{content})$ , with  $\beta$  **chosen on a validation split of users** (not on train).  $\beta = 0$  recovers SASRec, so fusion *cannot* do worse;  $\beta$  is selected for generalisation, not memorisation.

### 4. Evaluation protocol

We follow the dataset's **Global Temporal Split (GTS)**: 300 days train, a 30-min gap, a 1-day test window; user state frozen at the test start; users with no train history discarded. A **Listen+** positive is a play of  $\geq 50$  % of the track. We report **NDCG / Recall / Coverage @10 and @100**, ranking over the **full 629 k-item catalogue** (no sampled negatives) — so absolute values are small by design.

Two implementation notes that proved important: - **IDs are sparse** in a large space (uid up to  $1e6$ , item\_id up to  $9.39e6$  with only  $\sim 10$  k / 934 k present) → remapped to dense indices before any matrix / embedding. - **Seen items are never filtered.** Re-listening is real in music; filtering already-heard tracks collapses every model (MostPop 0.0171 → 0.0033, ItemKNN cosine 0.0418 → 0.0048).

**Harness validation.** Our numbers line up with the paper's, so comparisons are trustworthy:

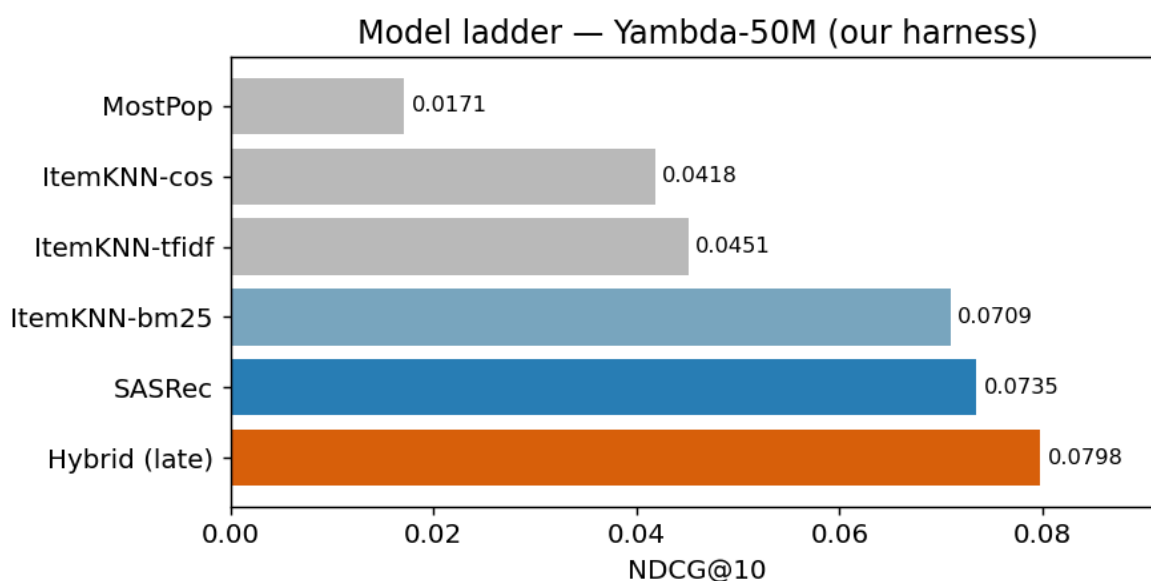
| Model          | NDCG@10 (ours) | NDCG@10 (paper) | ratio       |
|----------------|----------------|-----------------|-------------|
| MostPop        | 0.0171         | 0.0186          | 0.92        |
| ItemKNN (bm25) | 0.0709         | 0.0781          | 0.91        |
| SASRec         | 0.0735         | 0.0748          | <b>0.98</b> |

SASRec reproduces the paper to 98 %, so the harness  $\approx$  Yandex's; the  $\sim 9$  % gap on ItemKNN is undertuning (we used  $K = 100$ ), not a protocol difference.

## 5. Results

### 5.1 Model ladder (our harness, NDCG@10)

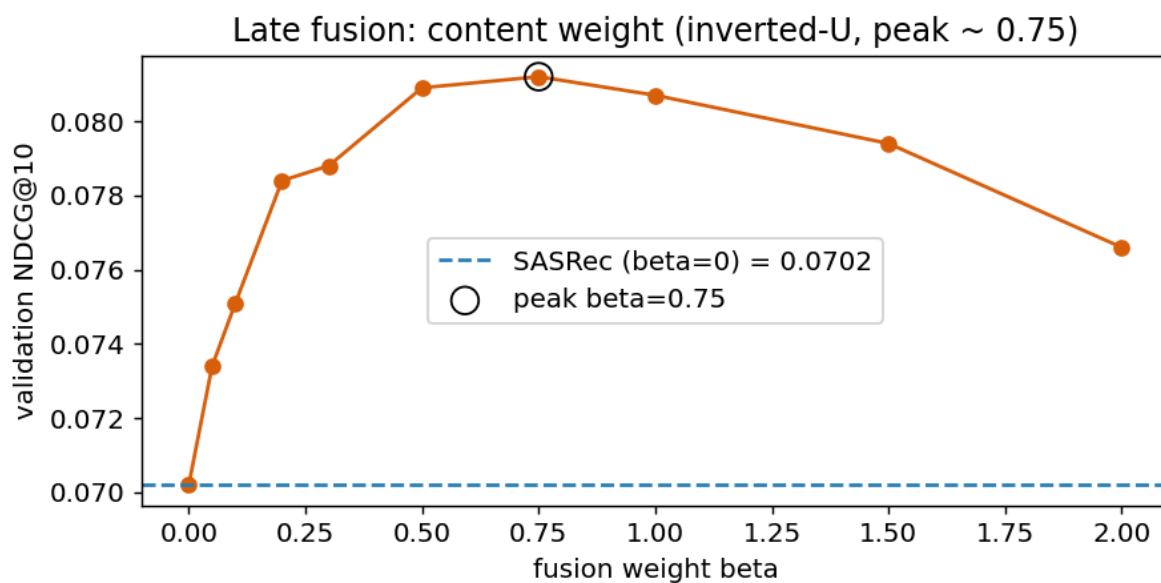
| Model                       | NDCG@10       | NDCG@100 | Recall@100 |
|-----------------------------|---------------|----------|------------|
| MostPop                     | 0.0171        | 0.0229   | 0.0353     |
| ItemKNN — cosine            | 0.0418        | 0.0738   | 0.1246     |
| ItemKNN — tfidf             | 0.0451        | 0.0786   | 0.1337     |
| ItemKNN — bm25              | 0.0709        | 0.1020   | 0.1609     |
| <b>SASRec</b>               | <b>0.0735</b> | 0.1005   | 0.1517     |
| Hybrid — joint fusion       | 0.0581        | 0.0908   | 0.1510     |
| <b>Hybrid — late fusion</b> | <b>0.0798</b> | —        | —          |



BM25 is the clear best classical weighting; SASRec beats all baselines.

### 5.2 Fusion weight $\beta$ (validation NDCG@10, one split)

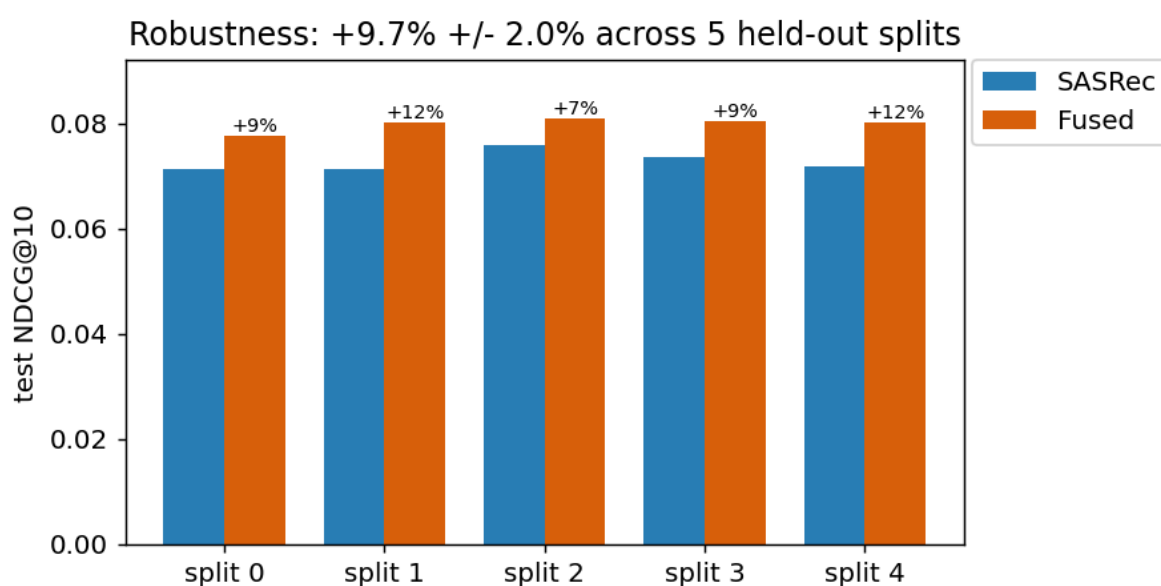
| $\beta$ | 0.0   | 0.05  | 0.1   | 0.2   | 0.3   | 0.5   | 0.75         | 1.0   | 1.5   | 2.0   |
|---------|-------|-------|-------|-------|-------|-------|--------------|-------|-------|-------|
| NDCG@10 | .0702 | .0734 | .0751 | .0784 | .0788 | .0809 | <b>.0812</b> | .0807 | .0794 | .0766 |



A clean inverted-U peaking at  $\beta \approx 0.75$  — the content adds genuine complementary signal (the optimum is not at  $\beta = 0$ ).

### 5.3 Robustness (5 held-out user splits)

| seed | $\beta$ | SASRec                                | Fused                                 | lift                                 |
|------|---------|---------------------------------------|---------------------------------------|--------------------------------------|
| 0    | 1.00    | 0.0712                                | 0.0776                                | +9.1 %                               |
| 1    | 0.75    | 0.0714                                | 0.0802                                | +12.3 %                              |
| 2    | 1.00    | 0.0759                                | 0.0809                                | +6.6 %                               |
| 3    | 0.75    | 0.0737                                | 0.0804                                | +9.1 %                               |
| 4    | 1.00    | 0.0718                                | 0.0801                                | +11.5 %                              |
| mean | —       | <b>0.0728 <math>\pm</math> 0.0018</b> | <b>0.0798 <math>\pm</math> 0.0011</b> | <b>+9.7 % <math>\pm</math> 2.0 %</b> |

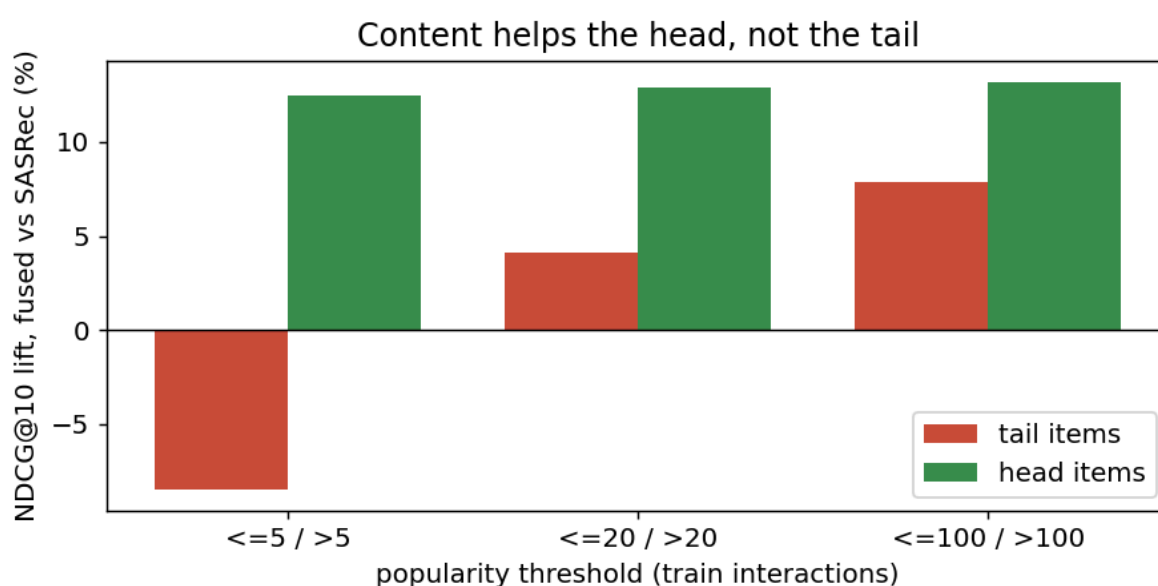


Every split is positive; the win is not an artefact of one split. On the held-out test users, all metrics improve (Recall@10 +11 %, NDCG@100 +11.5 %, Coverage@10 +15 %).

## 5.4 Where the gain comes from (tail analysis, $\beta = 0.75$ )

Each user's relevant test items are split by train popularity; recommendations are full-catalogue, we just credit hits per slice.

| slice           | users | SASRec | Fused  | lift    |
|-----------------|-------|--------|--------|---------|
| head > 5        | 4444  | 0.0691 | 0.0777 | +12.5 % |
| tail $\leq$ 5   | 3069  | 0.0134 | 0.0122 | -8.5 %  |
| head > 20       | 4256  | 0.0605 | 0.0683 | +12.9 % |
| tail $\leq$ 20  | 3853  | 0.0279 | 0.0290 | +4.1 %  |
| head > 100      | 3714  | 0.0467 | 0.0529 | +13.2 % |
| tail $\leq$ 100 | 4361  | 0.0471 | 0.0508 | +7.9 %  |



The audio gain concentrates on **popular** items (+12–13 %) and slightly **hurts** the extreme tail (−8.5 % on items with  $\leq 5$  interactions).

## 6. Discussion

**The audio signal is taste-alignment, not cold-start rescue.** Our user vector is the *mean* audio of the history — a taste centroid that boosts sonically-central (typically mainstream) tracks. It refines ranking on the head, but does not rescue rare items, and can even demote a genuine long-tail next-track by pulling mass toward audio-central items. This *overturns* the intuition in our pitch that content would fix cold-start, and is, we think, the more interesting finding: the benefit is real and robust, but its mechanism is different from the textbook story.

**Why late fusion beat joint fusion.** Joint training optimises a single train loss, so a high-capacity content branch is rewarded for memorising training transitions — it overfits. Late fusion selects the content weight by *validation* NDCG, so it adds content only to the extent it generalises ( $\beta \approx 0.75$ –1.0, never the overfit regime). The contrast — same audio signal, opposite outcome — is the core modelling lesson.

## 7. Reproducibility

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Everything is a pip-installable package ( `ymrec` ) driven by thin Kaggle notebooks that install it from GitHub. One free Kaggle T4 (16 GB) suffices; data and audio embeddings stream from Hugging Face in ~1–2 min.

| Notebook                         | Produces                              |
|----------------------------------|---------------------------------------|
| <code>00_kaggle_smoke</code>     | harness sanity check (MostPop)        |
| <code>01_baselines</code>        | MostPop + ItemKNN ladder              |
| <code>02_sasrec</code>           | SASRec (0.0735)                       |
| <code>03_content_emb_prep</code> | filtered audio embeddings (optional)  |
| <code>04_hybrid</code>           | joint-fusion negative result          |
| <code>05_fusion</code>           | late fusion + $\beta$ tuning          |
| <code>06_robustness</code>       | multi-seed robustness + tail analysis |
| <code>07_report_figures</code>   | the figures in this report            |

## 8. Conclusion & future work

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On real Yandex listening data we built a reproducible pipeline whose SASRec beats every published baseline (98 % of the paper) and whose late-fusion hybrid adds a **robust +9.7 %  $\pm$  2.0 %** NDCG@10 — backed by an honest account of *why*. Natural next steps: a **retrieval-oriented content design** (audio similarity to the most recent tracks rather than a global centroid) to target the long tail; richer fusion (per-user or per-cohort  $\beta$ ); and scaling to the 500M slice.

## References

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1. Kang & McAuley, *Self-Attentive Sequential Recommendation* (SASRec), ICDM 2018.
2. Yandex, *Yambda-5B — A Large-Scale Multi-modal Dataset for Ranking and Retrieval*, arXiv:2505.22238, RecSys 2025.